Kanav Chopra

AI Researcher | Full-Stack Developer | User-Centric Applications

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SUMMARY

Software Engineer with 5+ years of experience building scalable full-stack web applications. Experienced in React, TypeScript, Next.js, Node.js, and Python, with hands-on work on LLM-powered healthcare and enterprise systems, legacy modernization, and performance optimization.

TECHNICAL SKILLS

Frontend:	React.js, Next.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Vite.js, Vue.js
Tools & Infra:	Git, GitLab, CI/CD, Webpack, GraphQL, REST APIs, Server-Sent Events (SSE), WebSockets
Backend:	Python, FastAPI, Node.js, PostgreSQL, SQL, Docker, GCP, AWS
AI / ML:	LLM integration, Prompt Engineering, Generative AI, Agentic AI, Machine Learning, TensorFlow, Vector Embeddings, ReAct Paterns, Tool Orchestration, Evaluation Pipelines

EXPERIENCE

Stanford University — Palo Alto, CA
AI Researcher/Software Engineer — 06/2025 - Present

- Launched and led full stack development on SAGE (https://sage.arise-ai.org/), a decision-support tool for clinicians, cutting clinicians consulting time by **70-80%**. (from 15min to 2-3min)
- Engineered features for prompt versioning, feedback capture, and session tracking using React, TypeScript, Node.js, and Python, supporting experimentation with AI-assisted clinical notes
- Developed backend services to log user interactions, manage model prompt configurations, and facilitate feedback-driven iteration of AI responses
- Created Stanford Health Path (https://stanford-healthpath.vercel.app/),'Google Maps for patients to suggest healthcare routes using LLM.
- Developed and launched ARISE AI research network website using Next.js and a Headless WordPress CMS. (https://www.arise-ai.org/)

Vensure Employer Solutions — Noida, UP
Senior Software Engineer — 02/2024 - 08/2024

- Led modernization of legacy systems by transitioning to a microservices architecture, reducing technical debt by **40%** and mentored junior developers through code reviews and technical guidance.
- Helped select and implement modern technologies, improving system efficiency by **50%**.
- Designed and implemented scalable HR Management system features using React.js and TypeScript.
- Reviewed and refactored large legacy codebases, resolving critical issues and significantly reducing application crashes.

Ipsator Analytics — Bengaluru, KA
Senior Software Developer — 04/2023 - 12/2023

- Demonstrating expertise in designing visually appealing, intuitive, and user-friendly interfaces for web and mobile applications using React.js as a primary tech stack.
- Optimize UI performance by **60%** using minification, lazy loading, and reducing network requests.
- Built a scalable front-end architecture from scratch aligned with product requirements.

Khojdeal — Gurugram, Haryana

Software Developer — 07/2021 - 03/2023

- Developed multiple content-based website with complete web vitals optimization up to **100%.**
- Worked directly with marketing team to get better User matrix using Google Tag Manager(GTM) and Google Analytics(GA) integration resulting in **25%** improved Conversion rate.
- Deployed multiple websites using CI and CD pipeline with GitLab.
- Integrated cloud-flare with all organization and client running website to improve security and DNS management.

SVGS IT Solutions — Noida, UP

Software Developer — 09/2019 - 06/2021

- Developed application to access JSON and XML for RESTful web services using JavaScript.
- Designed full functional complex e-learning simulations using pure JavaScript and React.
- Converted Flash software applications into HTML, compatible with all mobile devices as well as browsers, resolving major browser compatibility issues.

PROJECTS

SAGE (https://sage.arise-ai.org/)

A secure, full-stack medical consultation and referral system that leverages real-time, multi-model AI orchestration to provide immediate, contextual clinical guidance.

- Architected a Multi-Model AI Orchestration System (FastAPI), integrating Claude, Gemini, and Azure OpenAI for dynamic, low-latency switching to optimize clinical accuracy and cost-efficiency.
- Engineered Real-Time, Contextual AI: Implemented a Server-Sent Events (SSE) pipeline with React/TypeScript and leveraged vector embeddings for high-accuracy clinical context matching, significantly reducing LLM hallucination.
- Ensured Auditable Healthcare Data Integrity: Integrated PostgreSQL for session management and implemented strict CORS and Pydantic validators for secure, protected handling of sensitive clinical data.
- Developed User-Centric Clinical Workflow: Built a modern, multi-step React/TypeScript/MUI interface with stringent input validation to enforce data quality and deliver a professional user experience for clinicians.

Clinical Research Multi-Agent System | Python, LangGraph, Vertex AII (clinical-research-agent)

- Built 5-agent LangGraph pipeline (planner, search, reader, critic, synthesizer) implementing ReAct, self-reflection, and hierarchical delegation across PubMed, Semantic Scholar, and arXiv.
- Swappable Claude/Gemini-on-Vertex providers via ADC; eval harness scoring citation accuracy, faithfulness, latency, and per-query token cost.

PUBLICATIONS

- 4 peer-reviewed publications (AMIA 2025, PSB / Biocomputing 2026) on LLM evaluation and AI-assisted clinical decision support — Google Scholar

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EDUCATION

Masters in Cybersecurity and Artificial Intelligence — *August 2024 – May 2026*

Webster University — *St. Louis, Missouri*

Bachelors in Computer Science — *August 2015 – May 2019*

University of Petroleum and Energy Studies — *Dehradun, UK*